

Math 2551 Worksheet - Review from Single-Variable Calculus and Linear Algebra

This worksheet reviews some concepts and tools of calculation from linear algebra, differential calculus, and integral calculus that will be useful for this course.

1. Let $\mathbf{v} = \langle 1, 2, 1 \rangle$ and $\mathbf{w} = \langle -1, 0, 2 \rangle$. Compute $2\mathbf{v}$, $\mathbf{v} - \mathbf{w}$, and $\mathbf{v} \cdot \mathbf{w}$.
2. Let a, b, c be variables. Calculate

$$\det \begin{bmatrix} a & b & c \\ 1 & -1 & 2 \\ 4 & 3 & -7 \end{bmatrix}$$

by expanding the determinant along the first row.

3. Draw the vectors $\mathbf{u} = 2\mathbf{i} + \mathbf{j}$ and $\mathbf{v} = -\mathbf{i} + 2\mathbf{j}$ starting at the origin in $\mathbb{R}^2$ to show that these vectors are orthogonal. Compute $\mathbf{u} \cdot \mathbf{v}$. What is the result?
4. Let $\mathbf{v} = \langle 2, -4, \sqrt{5} \rangle$ and $\mathbf{u} = \langle -2, 4, -\sqrt{5} \rangle$. Compute the following:
 - (a) $\mathbf{v} \cdot \mathbf{u}$
 - (b) the cosine of the angle θ between $\mathbf{v}$ and $\mathbf{u}$. Recall that $\mathbf{v} \cdot \mathbf{u} = |\mathbf{v}||\mathbf{u}| \cos(\theta)$
 - (c) $(3\mathbf{v}) \cdot (2\mathbf{u})$.
5. Compute the following integrals.

(a) $\int_0^\pi (\sin(\theta))^5 d\theta$

(c) $\int_1^2 t \ln(t) dt$

(b) $\int_0^\pi (\sin(\theta))^2 d\theta$

6. Find the maximum value of the function $f(t) = te^{t^2}$ on the interval from $t = 0$ to $t = 2$. Justify how you know this is the maximum.
7. Find the area enclosed by the curves $f(x) = x^2$ and $g(x) = 2 - x$.